

AYMANE AIT DADS

Audio Machine Learning Intern | Signal Processing · Deep Learning · Audio AI

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PROFESSIONAL SUMMARY

Built and deployed machine learning pipelines across audio, vision, and language domains, including an unsupervised transformer autoencoder for industrial anomaly sound detection achieving AUC 0.80 on unlabeled Mel spectrogram data. At Orange Maroc, designed a Random Forest + K-Means clustering pipeline on 100K+ network records that reduced manual analysis time by ~60% and mapped customer performance profiles to 5 commercial service tiers, delivering actionable maintenance recommendations via Power BI dashboards. Brings hands-on experience in PyTorch, deep learning model fine-tuning, feature engineering, and data-driven insight generation directly applicable to Bose audio AI research and product development workflows. Aims to apply signal processing, transformer architectures, and ML model evaluation expertise to advance Bose's next generation of audio machine learning systems.

CORE COMPETENCIES

Audio Machine Learning | Signal Processing | Deep Learning | Anomaly Detection | Transformer Architectures | Feature Engineering | Model Evaluation | Data Analysis

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed a **Random Forest + K-Means clustering pipeline** on 100K+ fiber-optic records (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Mapped customer performance profiles to **5 commercial service tiers** through automated feature engineering, with output adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** by automating end-to-end model evaluation and feature engineering workflows using Python, Scikit-learn, and Pandas.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, visualized through Power BI dashboards for non-technical audiences.

PROJECTS

Anomalous Sound Detection - Industrial Equipment

EURECOM Deep Learning · 2025 · AUC 0.80

- Designed an unsupervised **transformer autoencoder on Mel spectrograms** with SpecAugment for fault detection in slide-rail machinery with no labeled anomaly data, achieving AUC 0.80.
- Applied signal processing feature extraction and spectrogram-based deep learning to enable **predictive maintenance** entirely from unlabeled industrial audio recordings.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 CVPR · 2025 · Ranked 1st EURECOM Class

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark, fine-tuning CLIP ViT-L/14 (428M params) on 250K images.
- Built a custom MLP classification head (768 → 512 → 256 → 1, GELU, BatchNorm, Dropout) with dual learning rates and **10-view TTA ensemble** to maximize out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 · AdamW

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (Jun 2026) · \$106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · TRL · PEFT · vLLM · Triton

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Audio & Signal Processing: Mel Spectrograms | SpecAugment | Anomaly Detection | Transformer Autoencoder | Unsupervised Learning | Feature Engineering

ML / Deep Learning: PyTorch | Scikit-learn | Hugging Face Transformers | LoRA / PEFT | OpenCLIP | TRL | Unsloth

Data Science & MLOps: Python | Pandas | NumPy | Power BI | Git | FP16/BF16 Mixed Precision | Kaggle GPU